

Hollywood Crime Pattern Analysis Report

Nakul Bhatia, Justin Kim, Nikolai Cupid

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Introduction

The context of this project is that we are analyzing crime data in Hollywood, CA specifically from the years 2020 and 2024, as the goal of our project is to analyze how crime has changed in Hollywood, one of the most crime ridden areas of Los Angeles, since Covid, which began back in 2020.

Our research question was to analyze how crime patterns have changed since 2020. In order to achieve that, we took the crime data in Hollywood from just 2020 and 2024 and analyzed 3 different sub-goals: the ages of the victims of the crimes, the different types of crimes that occurred, and the monthly crimes trends in 2020 and 2024. We each then created tables and visualizations to further analyze these topics.

Data

Data Provenance

We got the data from Data.gov, and the data was originally provided by the Los Angeles Police Department through the City of Los Angeles ([Los Angeles](#)). The data consists of 28 attributes and 1,004,894 entries. The data is credible because it is published by the government and based on crime incident records from the Los Angeles Police Department.

One challenge that we had in this project was uploading the raw csv file into the repo. The size limit for file is 25 MB and the file was 240+ MB. So we had to use Git LFS to reference the csv file through a pointer in .gitattributes. This allowed us to put the file into the repo.

FAIR and CARE Principles

FAIR

The crime in Los Angeles data set satisfies the many FAIR principles. It's easily *findable* and *accessible* through Data.gov ([Los Angeles](#)). The data is easily *interpretable* as you can download the data in the CSV format with properly labeled columns and rows making it easy to read. The only issue is *reusability* as some cells were empty or had invalid entries (such as 0 for age).

CARE

The crime in Los Angeles data set does not single out any specific community/group of people. The data can be a *collective benefit* to these communities. But, others can use this data set to push harmful and damaging narratives about the communities who are affected by these crimes. Unfortunately, these communities do not have the *authority* to control how the data is used. We did not use the data in any such way and understand the *responsibility* and *ethics* we must have using this data.

Case and Attribute Description

The case in this data set is an individual crime incidents that occurred in 2020 and 2024. In order to answer our research question, we tidied the data set by selecting the 7 attributes from the original 28 attributes:

- ***Area***: Indicates where each crime incident occurred in Los Angeles. In this data set, are values are limited to '*Hollywood*'.
- ***Victim_Age***: Represents the age of victims and is used to categorize age groups.
- ***Victim_Sex***: Represents the gender of the victim.
- ***Year***: Indicates whether the incident occurred in 2020 or 2024.
- ***Month***: Represents the month of the incident with containing the values from 1 to 12.
- ***Time***: Represents the time of the incident with ranging from 00:00 to 23:59.
- ***crime_category***: Represents the type of each crime. The original 112 values were consolidated into four main crime category with an '*others*' category.

Exploratory Data Analysis

To improve accessibility, we used the ASU Image Accessibility Creator to help write descriptive alt text for our project visuals ([University](#)).

Victim Age Demographics

In this section, we want to explore and compare the amount victims in every age range from 2020 to 2024.

These ranges will go from 1-10, 11-20, and all the way until 91-100. We need to create separate tables for both 2020 and 2024. We want to create a neat tables that filter out every column from the original tidied table besides the year, the victim's age range, and create a new column with the amount of total victim's within that age range.

Table 1: Victim Age Range Distribution (2020)

Age Range	Number of Victims
1-10	30
11-20	345
21-30	2392
31-40	2095
41-50	1243
51-60	925
61-70	412
71-80	157
81-90	45
90-100	5

Table 2: Victim Age Range Distribution (2024)

Age Range	Number of Victims
1-10	5
11-20	129
21-30	1301
31-40	1261
41-50	658
51-60	397
61-70	209
71-80	94
81-90	20
90-100	6

Now, we will take the information from the above tables and use them to create visualizations.

We will create separate box plots for 2020 and 2024. The x-axis will have be the age ranges from 1-10 all the way to 91-100. The y-axis will be the amount of victims of crimes within that age range. The 2020 box plot is red and the 2024 box plot is blue.

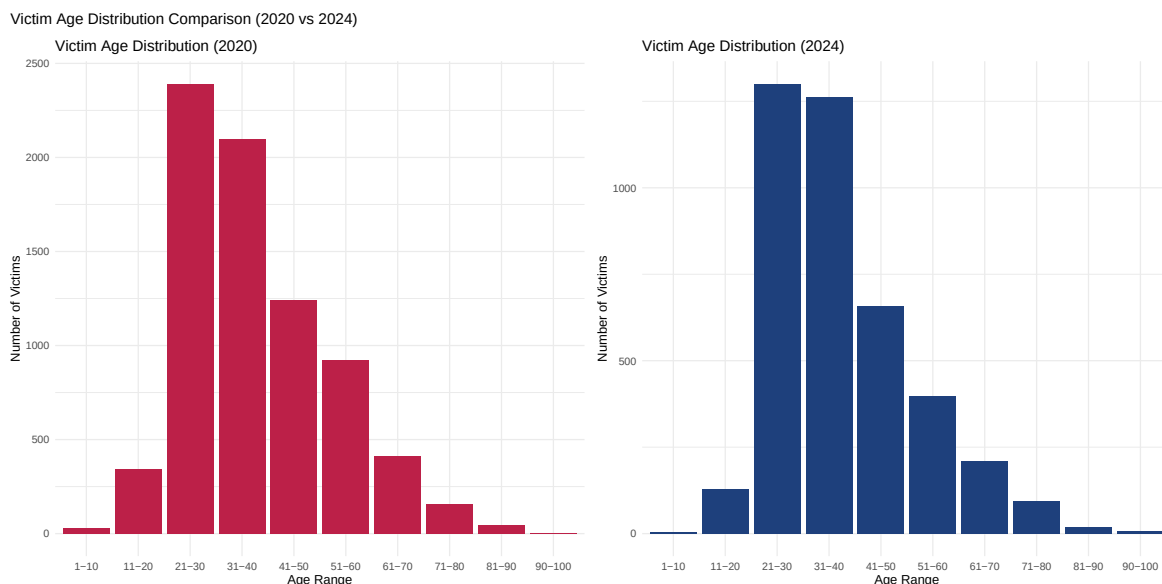


Figure 1: Victim Age Distribution Comparison (2020 vs 2024)

From: Table 1, Table 2, and Figure 1, The victim age distributions for 2020 and 2024 show a consistent pattern across both the tables and bar charts. In each year, the highest concentration of victims occurs in the 21–30 and 31–40 age ranges, indicating that young adults are the most affected group. Counts then steadily decline as age increases, with relatively few victims in both the youngest (1–10) and oldest (81+) groups.

From Table 1, Table 2, and Figure 1, Comparing the two years, there is a clear overall decrease in the number of victims in 2024 across every age range. This reduction is visible in both the tables (lower numerical values) and the bar charts (shorter bars). For example, the 21–30 and 31–40 groups—while still the largest—are significantly smaller in 2024 than in 2020.

Despite this decrease, the overall shape of the distribution remains unchanged. Both visualizations show a similar peak in young adulthood and a gradual decline toward older ages, suggesting that while the total number of victims has dropped, the relative distribution of age groups has remained stable over time.

As to why there is a general decrease from 2020 to 2024, we are unsure and require more research into that specifically.

Crime Category Trends

In this section, we created a table and data visualization to analyze and compare the types of crimes that occurred in 2020 and 2024.

For this subgoal, we focused on the attributes Year, Crime Category, and Count.

First, we used the tidied data set and transformed that into a clean and structured data summary table showing the crime trends in 2020 and 2024 over 5 different categories:

Table 3: Table of Crime Categories in Hollywood: 2020 vs 2024

Crime Category	2020 Crimes	2024 Crimes
Assault	1988	589
Burglary	1060	732
Other	1416	651
Robbery	343	82
Theft	2168	1661
Vandalism	674	365

Then we made a bar graph using the table we made previously:

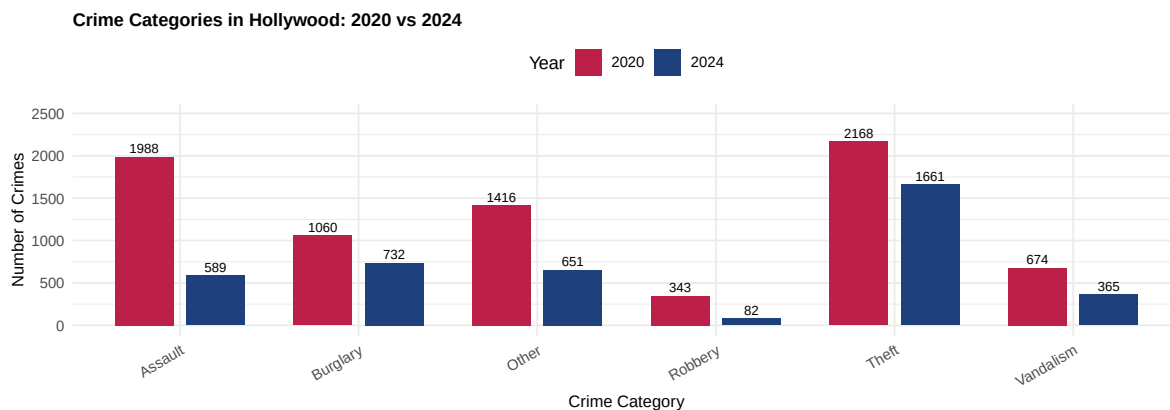


Figure 2: Crime Categories in Hollywood: 2020 vs 2024

Table 3 and Figure 2 show a clear trend in crime incidents in Hollywood between 2020 and 2024. Each visualization has its own strengths: the graph makes it easier to identify the trends, while the table provides precise crime counts in each crime category.

From Table 3, the table is divided into three main columns with the headings: “Crime Category”, “2020 Crimes”, and “2024 Crimes”. There are six rows beneath the headings, which each detail a different crime category and the number of crimes that are recorded in each year.

The categories listed are Assault, Burglary, Other, Robbery, Theft, and Vandalism. The numbers that are given for each category show a decline in all crime types from 2020 to 2024. In both 2020 and 2024, theft remains the most common crime in Hollywood and robbery remains the least common crime. There were 2168 counts of theft in 2020 and it decreased to 1661 in 2024. For robbery there were 343 counts of robbery and it decreased to only 82 in 2024.

From Figure 2, we can see that the image is a bar chart that compares statistics across the same 5 categories: Assault, Burglary, Other, Robbery, Theft, and Vandalism. The vertical axis represents the number of crimes, ranging from 0 to 2500, marked at intervals of 500. The horizontal axis lists the crime categories. Each category features two vertical bars: a red bar for 2020 and a blue bar for 2024. In the bar graph, the overall trend is more clearly visualized. We can see that red bars are higher than the blue bars in every single category. We can see that Assault, Other, Robbery, and Vandalism are all down at around 50% or more from 2020 to 2024. We can also see that Assault had the biggest decrease from 2020 to 2024 as the length of the bar has decreased significantly.

Monthly Crime Trends

In this section, we created data visualizations to analyze and compare monthly crime trends in Hollywood between 2020 and 2024.

In this subgoal, we focused on the attributes Year, Month, and crime_count.

First, the data were extracted from the tided data set and tranformed into a clean and well structured data summary table:

Table 4: Table of Monthly Crime Counts in Hollywood (2020 vs 2024)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2020	731	764	659	603	604	525	646	669	613	645	605	585
2024	665	622	644	305	246	252	228	247	244	269	207	151

Next, a line graph was created to represent the change of trend in crime counts of each year:

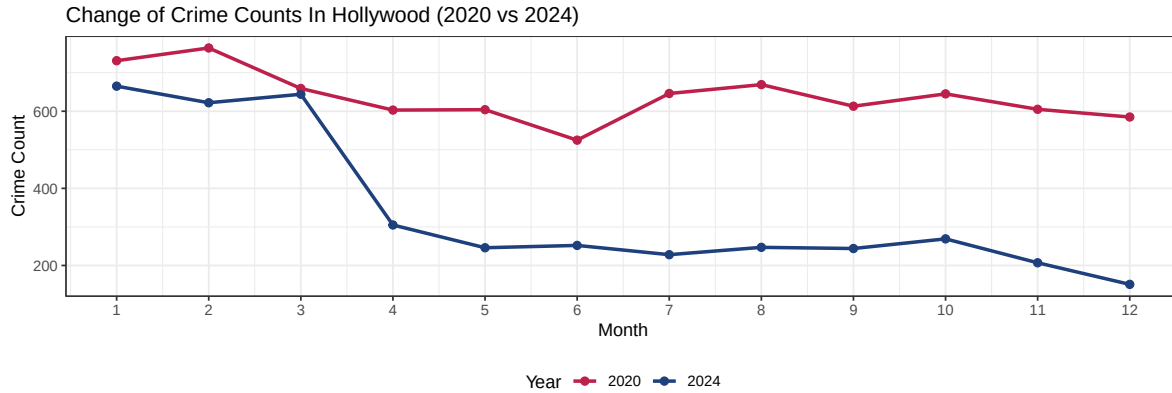


Figure 3: Graph of Monthly Crime Counts In Hollywood (2020 vs 2024)

Table 4 and Figure 3 show a clear trend in crime incidents in Hollywood between 2020 and 2024. Each visualization has its own strengths: the line graph makes it easier to identify the trends, while the summary table provides a precise monthly crime counts.

From Table 4, the crime incidents in both years were higher at the beginning of the year but declined toward the end of the year. For example, in 2020, there were 731 counts in January but decreased to 585 by December. A similar pattern was observed in 2024, where incidents dropped from 665 counts in January to 151 counts in December. Overall, the numbers of crime count for 2024 stayed relatively lower than 2020, with the closest values occurring in March, with 659 incidents in 2020 and 644 in 2024.

From Figure 3, the overall trend was more clearly visualized with the help of the line graph. For better clarification, the x-axis represents the months from 1 to 12 while the y-axis represents the crime count, ranging from 100 to 800. In 2020, the trend remained relatively stable with moderate fluctuations throughout the year, with the lowest crime count shown in June. In contrast, 2024 showed a sharper and dramatic decline in the early months, particularly between March and April, before remaining relatively stable decrease for the remainder of the year.

In addition, both years showed a general decreasing trend in crime incidents as the months progressed. One notable insight from the data was a slight increase in August and October for both years, with a decline again afterward. This pattern may be related to seasonal events, such as summer break in August and Halloween in October, which may contributed to increased crime activity in a major tourist area like Hollywood.

Overall, the results indicate that crime counts in 2024 remained relatively lower than those in 2020.

Conclusion

We observed that the crime trends in 2024 were overall lower than those in 2020 in the monthly trend and the overall crime type trend. This finding was unexpected because we initially assumed crime levels would be lower in 2020 due to the COVID-19 pandemic as people were limited to do outside activities and people were in their homes and quarantined. There may be several possible reasons for this decline, such as improvements in security due to advanced technology or change in social behavior after the pandemic. Therefore, further analysis would be required to find out the exact causes, suggesting a possible area for future research.

For the victim's age demographics, we found that a lot of crimes were committed against those from ages 21-40. The general trend of number of victims stayed consistent between both plots but we still see that the number of victims in 2020 is noticeably higher than 2024. For the crime categories, certain categories like Assault and Robbery dropped significantly percentage wise from 2020 to 2024 in comparison to other categories like Theft and Burglary. There were most likely certain factors that caused a higher drop in crime count for some categories, that would suggest another possible area of future research.

Author Contribution

Justin: README.md, Plan.md, Creating Images, raw_data, and project_code folders, Visualization and narration for subgoal #3, Conclusion, Case and Attribute Description, and Finding the data set online.

Nakul: Main author of the Quarto file, Author of Introduction, Visualization and narration for subgoal #2, Conclusion, Data Provenance, and wrote crime_tidy_dataset.R.

Nikolai: FAIR and CARE Principles, Visualization and narration for subgoal #1, Conclusion, and Loaded large CSV file into repo using GIT LFS.

References

Los Angeles, City of. *Crime Data from 2020 to 2024*. Data.gov, Feb. 2020, https://catalog.data.gov/dataset/crime-data-from-2020-to-present?from_hint=eyJzb3J0IjoicG9wdWxhcml0eSJ9.
University, Arizona State. *Image Accessibility Creator*. ASU Online AI Labs, https://asuo-ai-labs.streamlit.app/Image_Accessibility. Accessed 4 May 2026.

Code Appendix

```
# Style Guide: Tidyverse Style Guide
# Main Author: Nakul Bhatia
# Reviewed by: Justin Kim, Nikolai Cupid

library(tidyverse)
library(knitr)
library(kableExtra)
library(ggplot2)
library(dplyr)
library(readr)
library(lubridate)
library(patchwork)
library(tidyr)
library(scales)
library(janitor)
# Los Angeles Crime Data Tidying
## Import and clean 2020-2025 Los Angeles crime data
## Prepare data for crime trend analysis.
## Focus on month, type, age, and sex
# Style Guide: Tidyverse Style Guide
# Primary Author: Nakul Bhatia
# Reviewed by: Justin Kim

# Step 1: Load packages needed for data wrangling

# Step 2: Import the raw crime data set and view it
crimeRaw <- read.csv("raw_data/Crime_Data_from_2020_to_2024.csv")

# View(crimeRaw)

# Step 3: Select only the variables needed for this project
crimeTidy <- crimeRaw %>%
  select(
    DATE.OCC, AREA.NAME, Crm.Cd.Desc, Vict.Age, Vict.Sex
  )

# Step 4: Filter the data set to only include crimes that occurred in Hollywood
crimeHollywood <- crimeTidy %>%
  filter(AREA.NAME == "Hollywood")
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# Step 5: Convert the date column to date-time format
# Keep only entries from 2020 and 2024
crimeHollywood$DATE.OCC <- as.POSIXct(
  crimeHollywood$DATE.OCC,
  format = "%m/%d/%Y %I:%M:%S %p"
)
crimeHollywood_filtered <- crimeHollywood %>%
  filter(format(DATE.OCC, "%Y") %in% c("2020", "2024"))

# Step 6: Clean the data by removing incomplete or invalid victim information
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  filter(
    complete.cases(.),          # removes any NA
    Vict.Age != 0,              # remove age = 0
    Vict.Sex != "X",           # remove invalid sex
    Vict.Age != "",            # remove empty strings (if stored as text)
    Vict.Sex != ""
  )

# Step 7: Check the unique crime descriptions
# unique(crimeHollywood_filtered$Crm.Cd.Desc)

# Step 8: Create separate year and month columns from the date column
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  mutate(
    Year = year(DATE.OCC),
    Month = month(DATE.OCC)
  )

# Step 9: Remove the original date column
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  select(-DATE.OCC)

# Step 10: Rename columns to make the data set more readable
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  rename(
    Area = AREA.NAME,
    Crime_Type = Crm.Cd.Desc,
    Victim_Age = Vict.Age,
    Victim_Sex = Vict.Sex
  )

```

```

# Step 11: Group detailed crime descriptions into broader crime categories
# Create a separate column for the broader categories
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  mutate(crime_category = case_when(
    grepl("BURGLARY", Crime_Type) ~ "Burglary",
    grepl("THEFT", Crime_Type) ~ "Theft",
    grepl("ASSAULT", Crime_Type) ~ "Assault",
    grepl("ROBBERY", Crime_Type) ~ "Robbery",
    grepl("VANDALISM", Crime_Type) ~ "Vandalism",
    TRUE ~ "Other"
  ))

# Step 12: Remove the detailed crime type column after creating the broader category
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  select(-Crime_Type)

# Step 13: Rename the new crime category column to keep consistent with other columns
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  rename(Crime_Category = crime_category)

# Step 14: Keep only male and female victim record and recode the labels for readability
crimeHollywood_filtered <- crimeHollywood_filtered %>%
  filter(Victim_Sex %in% c("M", "F")) %>%
  mutate(
    Victim_Sex = recode(Victim_Sex,
                        "M" = "Male",
                        "F" = "Female")
  )

# Victim Age Demographic Tables in Hollywood
## Create victims per age range (0-10, 11-20, etc.) tables for 2020 and 2024
# Style Guide: Tidyverse Style Guide
# Primary Author: Nikolai Cupid
# Reviewed by: Nakul Bhatia

#Goal 1: Create 2 separate tables for 2020 and 2024. These tables will
#include both age ranges (1-10, 11-20, etc.) and the number of victims per
#that age range

#Step 1: Create a table that changes the age column into the age range it's in
crimeHollywood_ages_filtered <- crimeHollywood_filtered %>%
  mutate(Age_Range = cut(
    Victim_Age,

```

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#Step 1a: splits age into bins of 0-10, 11-20, all the way to 100
breaks = seq(0, 100, by = 10),
#Step 1b: assigns the readable table values
labels = c("1-10", "11-20", "21-30", "31-40", "41-50",
           "51-60", "61-70", "71-80", "81-90", "90-100"),
include.lowest = TRUE
))

#Step 2: creates 2020 table with age range and number of victims in that age range
age_table_2020 <- crimeHollywood_ages_filtered %>%
  filter(Year == 2020) %>%
  group_by(Age_Range) %>%
  summarise(Victim_Count = n())

#Step 3: creates a neater version of the above table using kableExtra
crime_age_table_2020 <- age_table_2020 %>%
  kable(
    #Step 3a: Gives the table a header
    col.names = c("Age Range", "Number of Victims"),
    align = "lr",
    row.names = FALSE,
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = "scale_down",
    font_size = 9
  )

#Step 4: Display the tables
crime_age_table_2020

#Step 5: creates 2024 table with age range and number of victims in that age range
age_table_2024 <- crimeHollywood_ages_filtered %>%
  filter(Year == 2024) %>%
  group_by(Age_Range) %>%
  summarise(Victim_Count = n())

#Step 6: creates a neater version of the above table using kableExtra
crime_age_table_2024 <- age_table_2024 %>%
  kable(
    col.names = c("Age Range", "Number of Victims"),
    align = "lr",
    row.names = FALSE,

```

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    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options = "scale_down",
    font_size = 9
  )

#Step 7: Display the tables
crime_age_table_2024
# Victim Age Demographic Plots in Hollywood
## ## Create victim age-range visualizations for 2020 and 2024
# Style Guide: Tidyverse Style Guide
# Primary Author: Nikolai Cupid
# Reviewed by: Justin Kim

#Goal 2: Create 2 separate bar charts for 2020 and 2024. The x axis will be
#the age ranges while the y axis will be the number of victims in that age range

#Step 1: Create a bar chart for the 2020 victim age distribution

#Step 1a: Create a plot with the x axis being the age range and y axis being the
#number of victims
crime_age_plot_2020 <- ggplot(
  age_table_2020, aes(x = Age_Range, y = Victim_Count)) +
  #Step 1b: Creates the bars from the values in victim_count
  geom_col(fill = c("#BC2048")) +
  #Step 1c: Creates the title and renames the x and y axes
  labs(
    title = "Victim Age Distribution (2020)",
    x = "Age Range",
    y = "Number of Victims"
  ) +
  theme_minimal()

#Step 2: Create a bar chart for the 2024 victim age distribution following same
#steps as above for Step 1
crime_age_plot_2024 <- ggplot(
  age_table_2024, aes(x = Age_Range, y = Victim_Count)) +
  geom_col(fill = c("#1E407C")) +
  labs(
    title = "Victim Age Distribution (2024)",
    x = "Age Range",

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    y = "Number of Victims"
  ) +
  theme_minimal()

crime_age_plot_2020 + crime_age_plot_2024 +
  plot_annotation(
    title = "Victim Age Distribution Comparison (2020 vs 2024)"
  )
# Style Guide: Tidyverse Style Guide
# Primary Author: Nakul Bhatia
# Reviewed by: Nikolai Cupid

# Step 1: Create a long-format summary table for the bar graph

# Step 1a: Start with the cleaned Hollywood crime data set
crime_type_graph_data <- crimeHollywood_filtered %>%
  # Step 1b: Group the data by year and crime category
  group_by(Year, Crime_Category) %>%
  # Step 1c: Count the number of crimes in each year/category group
  summarise(Count = n(), .groups = "drop")

# Step 2: Create a wide-format summary table for viewing crime categories by year

# Step 2a: Start with the cleaned Hollywood crime data set
crime_type_table_wide <- crimeHollywood_filtered %>%
  # Step 2b: Group the data by crime category and year
  group_by(Crime_Category, Year) %>%
  # Step 2c: Count the number of crimes in each crime category
  summarise(Count = n(), .groups = "drop") %>%
  # Step 2d: Reshape the table so 2020 and 2024 become separate columns
  pivot_wider(
    names_from = Year,
    values_from = Count,
    values_fill = 0
  ) %>%
  # Step 2e: Rename columns and organize rows alphabetically by crime category
  rename(
    `Crime_Category` = Crime_Category,
    `2020 Crimes` = `2020`,
    `2024 Crimes` = `2024`
  ) %>%
  arrange(`Crime_Category`)

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# Step 3: Create a formatted table with title, centered number column, and striped styling

# Step 3a: Start with the wide-format table created above
crime_type_table_formatted <- crime_type_table_wide %>%
  # Step 3b: Use kable() to add a caption, align columns, and remove row names
  kable(
    align = "lcc",
    row.names = FALSE,
    booktabs = TRUE
  )

# Step 4: Display the formatted table
crime_type_table_formatted
# Style Guide: Tidyverse Style Guide
# Primary Author: Nakul Bhatia
# Reviewed by: Justin Kim

# Step 5: Create a grouped bar graph comparing category counts in 2020 and 2024

# Step 5a: Start the plot using the long-format graph data
crime_category_bar <- ggplot(
  crime_type_graph_data,
  # Step 5b: Set crime category on x-axis, count on y-axis, and year as fill color
  aes(x = Crime_Category, y = Count, fill = factor(Year))
) +
  # Step 5c: Create side-by-side bars for 2020 and 2024 within each crime category
  geom_bar(stat = "identity",
    position = position_dodge(width = 0.8),
    width = 0.65
  ) +
  # Step 5d: Add the count labels above each bar
  geom_text(
    aes(label = Count),
    position = position_dodge(width = 0.8),
    vjust = -0.4,
    size = 2.8
  ) +
  # Step 5e: Add extra space above the tallest bar so the labels are not cut off
  expand_limits(y = max(crime_type_graph_data$Count) * 1.15) +
  # Step 5f: Add the graph title, axis labels, and legend title
  labs(

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    title = "Crime Categories in Hollywood: 2020 vs 2024",
    x = "Crime Category",
    y = "Number of Crimes",
    fill = "Year"
  ) +
  # Step 5g: Manually set the colors for 2020 and 2024
  scale_fill_manual(
    values = c("2020" = "#BC2048", "2024" = "#1E407C")
  ) +
  # Step 5h: Use a clean minimal theme
  theme_minimal() +
  # Step 5i: Adjust title, axis text, axis labels, and legend placement
  theme(
    plot.title = element_text(hjust = 0, size = 11, face = "bold"),
    axis.text.x = element_text(angle = 30, hjust = 1, size = 9),
    axis.title.x = element_text(size = 10),
    axis.title.y = element_text(size = 10),
    legend.position = "top"
  )

# Step 6: Display the grouped bar graph
crime_category_bar
## Goal 1: Create a clean data table that shows the number of monthly crime
## incidents in Hollywood for 2020 and 2024.

#Step 1: Create a summary data set of monthly crime count from the tidied crime data set.
crime_summary <- crimeHollywood_filtered %>%
##Step 1a: Group the data by year and month using group_by().
  group_by(Year, Month) %>%
##Step 1b: Count total number of crime incidents in each group and then remove grouping
  summarise(crime_count = n(), .groups = "drop") %>%
##Step 1c: Use arrange() to sort the date in chronological order by year and month.
  arrange(Year, Month)

#Step 2: Change the table from long to wide format.
crime_summary_wide <- crime_summary %>%
##Step 2a: Use month.abb to change the integer into characters. (e.g. 1 -> Jan)
  mutate(Month = factor(month.abb[Month])) %>%
##Step 2b: Use pivot_wider to transform the table into wide format.
  pivot_wider(
    names_from = Month,
    values_from = crime_count
  )

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)

#Step 3: Create neat summary table from monthly crime summary data set
crime_month_table <- crime_summary_wide %>%
##Step 3a: Use kable() to create caption and set column for the table.
  kable(
    align = "lrrrrrrrrrrrrr",
    row.names = FALSE,
    booktabs = TRUE
  )

crime_month_table
##Goal 2: Create a line graph that analyzes and compares monthly crime trends
## in Hollywood between 2020 and 2024.

#Step 1: Create the framework and specify the mappings.
crime_month_plot <- ggplot(
  data = crime_summary,
  mapping = aes(
    x = Month,
    y = crime_count,
    color = factor(Year)
  )
) +
#Step 2: Use geom_point() to create a point of the count and geom_line()
#to connect the point.
  geom_point(size = 2) +
  geom_line(linewidth = 1) +
##Step 2a: Add axis labels, legend title, and graph title.
  labs(
    x = "Month",
    y = "Crime Count",
    color = "Year",
    title = "Change of Crime Counts In Hollywood (2020 vs 2024)"
  ) +
##Step 2b: Set the scale for x-axis from 1 to 12 to represent each month of the year.
  scale_x_continuous(
    limits = c(1, 12),
    breaks = 1:12
  ) +

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#Step 3: Use scale_color_manual() to assign different colors to each year.
scale_color_manual(
  values = c(
    "#BC204B", "#1E407C"
  )
) +
#Step 4: Add a black and white background with legends placed at the bottom of the graph.
theme_bw() +
theme(legend.position = "bottom")

#Step 5: Check and polish the code using PCIP method.

crime_month_plot
```